MultiWikiQA: A Reading Comprehension Benchmark in 300+ Languages

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Abstract

We introduce a new reading comprehension dataset, dubbed MultiWikiQA, which covers 306 languages. The context data comes from Wikipedia articles, with questions generated by an LLM and the answers appearing verbatim in the Wikipedia articles. We conduct a crowdsourced human evaluation of the fluency of the generated questions across 30 of the languages, providing evidence that the questions are of good quality. We evaluate 6 different language models, both decoder and encoder models of varying sizes, showing that the benchmark is sufficiently difficult and that there is a large performance discrepancy amongst the languages. The dataset and survey evaluations are freely available.

1. Introduction

Extracting information from documents is one of the primary uses of large language models (LLMs), especially with the rise of retrieval-augmented generation (RAG) use cases. Reading comprehension, also known as extractive question answering, is a key component in such information extraction. At its core, it concerns locating an answer to the user's query within the provided document.

This relevance of reading comprehension tasks to downstream use cases also increases the importance of having access to high-quality reading comprehension evaluation datasets within all languages.

In this work, we generate a reading comprehension dataset for 306 different languages, based on Wikipedia articles, thus increasing the access to evaluation datasets within all of these languages. The questions are all generated with an LLM, and we evaluate the quality of the generated questions within 30+ of the languages through crowdsourcing. Lastly, we evaluate several language models on all of the languages, mapping out the performance of these models across a wide variety of languages. Our key contributions are:

1. Release of a multilingual reading comprehension dataset in 306 languages for evaluation of encoder, decoder and encoder-decoder language models.[1].
2. Results and raw data from 156 crowdsourced quality evaluations of the LLM-generated questions within the dataset, across 30 languages.[2].
3. Evaluations of 6 different language models on 261 languages.

2. Related Work

Many reading comprehension datasets have been published in different languages, including English (**???**), Polish (**?**), Korean (**?**), Norwegian (**??**), German (**?**), French (**?**), Icelandic (**?**), Faroese (**?**) and Russian (**?**). Multilingual reading comprehension datasets have also been released, covering 7 languages (**?**), 26 languages (**?**), 11 languages (**?**), and 3 languages (Nielsen, 2023).

All of these benchmarks only cover a small fraction of the world's written languages, leaving most of the low-resource languages behind. Belebele (**?**) is a notable exception, which spans an impressive 122 languages. The multiple-choice format of Belebele is quite different compared to regular extractive question answering datasets, however. Furthermore, it is abstractive, contains only 900 samples for each language, and these samples only have short contexts of approximately 500 characters.

3. Methodology

3.1. Dataset Generation

The dataset generation methodology closely follows the methodology in **?**, albeit with minor tweaks and using a different LLM, which we will describe below - it is also illustrated in Figure 1.

From a given document, we start by generating tentative questions and answers with the LLM using the system and user prompt in Figure 2. We ask the model to generate 2-10 different question-answer pairs for each article, both for efficiency and diversity reasons, since it is more likely to generate new question-answer pairs when conditioned on

[1]The dataset can be found at https://huggingface.co/datasets/alexandrainst/multi-wiki-qa.

[2]The raw quality evaluation data can be found at [redacted].

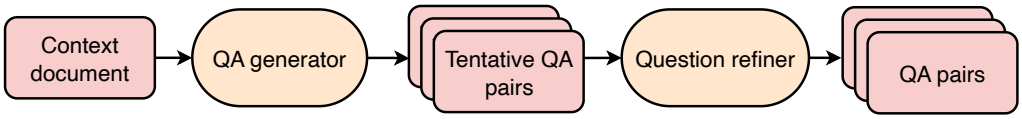

Figure 1: MultiWikiQA dataset generation process.

```
SYSTEM:
You are a helpful {language} question answering
dataset generator. The only language you know is
{language}.

USER:
The following is a Wikipedia article in {language}.

<article>
{article}
</article>

Generate 2 to 10 questions about the article,
depending on the length of the article, all
of which answered in the article.

You also have to supply answers to the questions,
and the answers have to appear exactly as written
in the article (including same casing).

The answers should only contain the answers
themselves, and not the surrounding sentence -
keep the answers as short as possible.

The answers have to be different from each other.

All your questions and answers must be in
{language}.

Your answer must be a JSON dictionary with the
key "results", with the value being a list of
dictionaries having keys "question" and "answer".
```

Figure 2: The system and user prompt used to generate the tentative questions and answers.

the previously generated pairs. We use structured generation to ensure a valid JSON output.

Next, we filter the generated JSON dictionaries by checking that each question-answer entry contains the appropriate "question" and "answer" keys, as well as checking if the answer appears verbatim in the context document.

We *could* stop at this point, as we now have a set of questions and answers for the context document. However, other reading comprehension datasets have been criticised for having questions that used the same wording as the context document, making it too easy for language models to "cheat" by simply word matching (Weissenborn et al., 2017; Jia and Liang, 2017). In an attempt to prevent this, we proceed with a separate rephrasing stage, where we prompt the same LLM to rephrase the question (without the context), using the prompt in Figure 3.

The resulting set of context-question-answer triples are then collected into a dataset of the same format as SQuAD (**?**).

```
The following is a {language} question.

<question>
{question}
</question>

Re-write the question as much as possible,
preserving the meaning, using synonyms,
other phrases, or a different (valid)
word order.

Your question must be in {language}.

Your answer must be a JSON dictionary with the
key "question".
```

Figure 3: The prompt used to rephrase the generated tentative questions.

3.2. Quality Evaluation of LLM-generated Questions

To evaluate the quality of the LLM-generated questions, we conducted a survey in all the included languages, and crowdsourced answers from various social media channels. Each survey contained a random sample of 50 generated questions for the given language, and prompted the user to rate the fluency of each question as 1, 2, or 3 stars. The precise preamble that was presented to each user can be seen in Figure 4.

We used the Microsoft Forms service[3] to facilitate the individual language surveys, and we self-hosted a simple routing interface which guided users to the correct language survey - the interface can be seen in the appendix. As each survey was created manually, we did not create the surveys for all languages, but instead had the routing interface send us an email if the user selected a language not currently covered - we then sent an email reply to the user when the language was included. The routing interface was coded using Vue.js (You, 2025) - the source code can be found in the appendix.

4. The Dataset

We generate the dataset using the methodology in Section 3.1. We use the 20231101 Wikipedia dump[4] and include 315 of the Wikipedia languages - a full list can be found in the appendix. We include special cases for Mandarin and Portuguese.

[3] https://forms.cloud.microsoft/

[4] https://hf.co/datasets/wikimedia/wikipedia

Please indicate how natural the following questions are:

⋆ *Does not sound natural at all*

⋆⋆ *Sounds mostly natural, but there is a particular part of the question that looks wrong*

⋆⋆⋆ *Sounds like a natural question*

Note that "naturalness" here is only meaning fluency, so whether the question is unanswerable or requires context to be answered does not matter here.

Figure 4: The preamble used in all of the surveys.

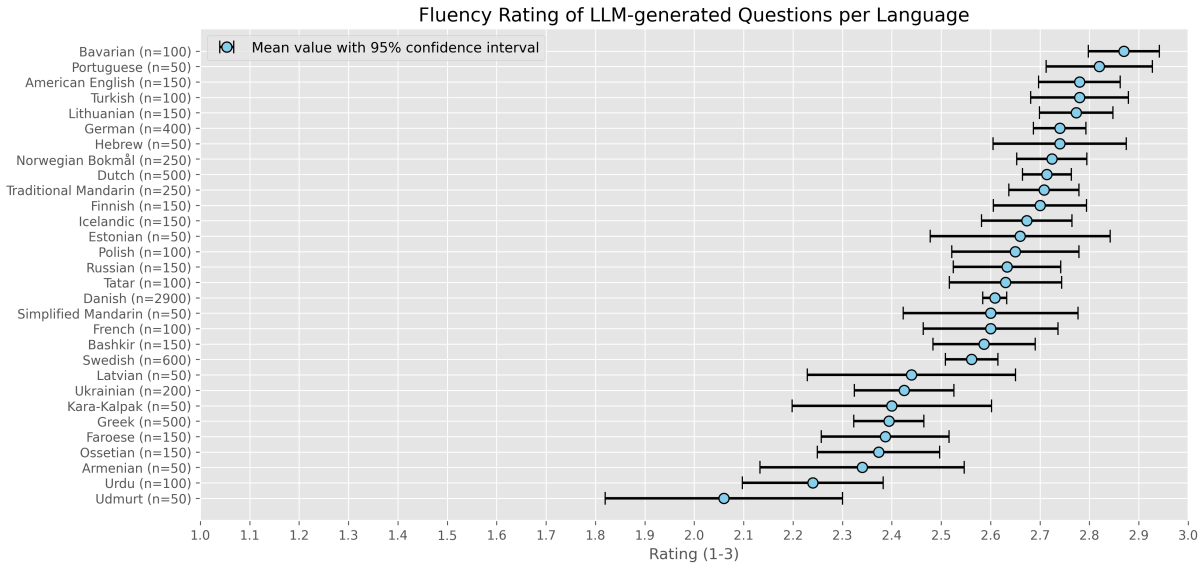

Figure 5: Results from the conducted fluency surveys.

We split Mandarin articles into Simplified Mandarin ("zh-cn") and Traditional Mandarin ("zh-tw") using the HanzIdentifier (tsroten, 2024), and we split Portuguese articles into European Portuguese ("pt-pt") and Brazilian Portuguese ("pt-br") using the PtVid classifier (**?**).

We use the Gemini-1.5-pro model (Reid et al., 2024) for the question generation with temperature 1.0 and where we allow 1,000 generated tokens, and we stop generating for a given language when we reach 5,000 context-question-answer samples, or when we run out of articles. We ran out of articles for 101 languages - see the appendix for an overview of these languages.

Using the question evaluation methodology in Section 3.2, we get 156 survey responses in 30 different languages. The mean quality scores across the languages, along with the number of survey responses, can be found in Figure 5. We see that the generated questions have a mean rating above 2.0, corresponding to "mostly natural", even for the languages Bashkir, Kara-Kalpak, Faroese, Ossetian, Udmurt and Icelandic, all having fewer than one million native speakers.

5. Evaluations

We evaluate a variety of language models on the MultiWikiQA dataset in all the languages with at least 1,024 samples for training, 32 for validation and 128 for testing. This was chosen as we are also evaluating encoder models, and 1,024 samples was found in Nielsen (2023) to be enough for the models to adequately fit the data for several reading comprehension datasets and languages. There were 264 languages satisfying this criterion.

The evaluation itself was conducted using the EuroEval framework (Nielsen, 2023; Saattrup Nielsen et al., 2025). See the list of evaluated models in Table 1. The decoder models were evaluated 2-shot, which was preferred over zero-shot evaluation to enable proper evaluation of base decoder models. The few-shot examples come from the training split. The encoder models were trained on the training split, with early stopping based on the validation split, and the final performance reported on the test split.

The results are visualised in Figure 6, and the full results can be found in the appendix. From the results, we see that there is a discrepancy in performance across languages, which is consistent across the three different model types. We also

Model Name	Parameters	Type	Mean F1-score
Mistral-Small-3.1-24B-Instruct-2503 (Mistral-AI, 2025)	24B	Instruct Decoder	55.83% ± 1.09%
Mistral-Small-3.1-24B-Base-2503 (Mistral-AI, 2025)	24B	Base Decoder	54.71% ± 1.20%
Llama-3.1-8B-Instruct (Grattafiori et al., 2024)	8B	Instruct Decoder	52.38% ± 0.91%
Llama-3.1-8B (Grattafiori et al., 2024)	8B	Base Decoder	47.26% ± 1.22%
Multilingual-E5-large (Wang et al., 2024)	560M	Encoder	23.82% ± 0.65%
XLM-RoBERTa-large (Ruder et al., 2019)	561M	Encoder	20.23% ± 0.69%

Table 1: Evaluation results on MultiWikiQA in 261 languages

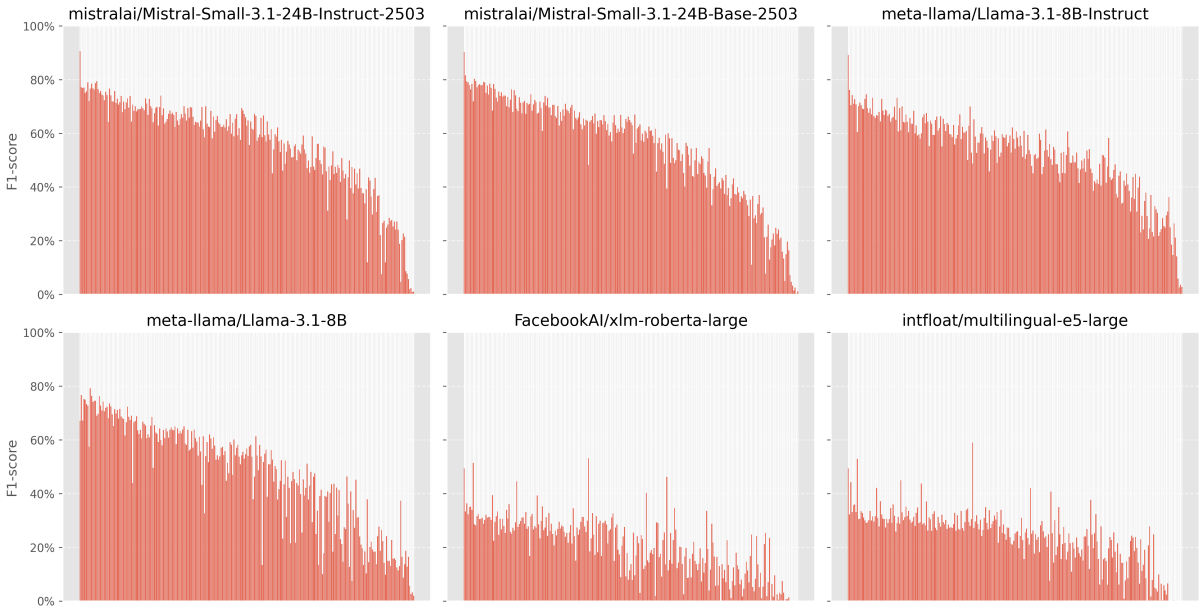

Figure 6: F1-score performance of six models across multiple languages, where the languages on the x-axis are sorted in descending order based on the mean F1-score across all models.

see that the task is not saturated for any of the languages, and with it being especially challenging for many of them.

6. Conclusion

We have introduced a new reading comprehension dataset in 306 languages, based on Wikipedia articles, with questions generated by an LLM. We crowdsourced the quality of the LLM-generated questions for 30 languages, showing that the LLM-generated questions are of good quality. Lastly, we evaluated 6 models on all of the languages, which showed that the benchmark is sufficiently difficult across instruction-tuned decoders, base decoders and encoders, and that there is a large performance discrepancy amongst the languages.

Limitations

While we got survey responses in 30 different languages, that still only covers approximately 10% of the languages covered in the dataset, so we cannot guarantee that the conclusions from the surveys generalise to the remaining languages. However, since the surveys cover a wide spectrum of language families and language resource levels, we are quite confident in such a generalisation.

Acknowledgements

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7. Bibliographical References

Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, Marcel Blistein, Ori Ram, Dan Zhang, Evan Rosen, et al. 2025. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Robin Jia and Percy Liang. 2017. Adversarial examples for evaluating reading comprehension systems. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pages 2021–2031.

Mistral-AI. 2025. Mistral small 3.1. [Online; accessed 31. Jul. 2025].

Dan Nielsen. 2023. ScandEval: A benchmark for Scandinavian natural language processing. In *Proceedings of the 24th Nordic Conference on Computational Linguistics (NoDaLiDa)*, pages 185–201, Tórshavn, Faroe Islands. University of Tartu Library.

Machel Reid, Nikolay Savinov, Denis Teplyashin, Dmitry Lepikhin, Timothy P Lillicrap, Jean-Baptiste Alayrac, Radu Soricut, Angeliki Lazaridou, Orhan Firat, Julian Schrittwieser, et al. 2024. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *CoRR*.

Sebastian Ruder, Anders Søgaard, and Ivan Vulić. 2019. Unsupervised cross-lingual representation learning. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics: Tutorial Abstracts*, pages 31–38.

Dan Saattrup Nielsen, Kenneth Enevoldsen, and Peter Schneider-Kamp. 2025. Encoder vs decoder: Comparative analysis of encoder and decoder language models on multilingual NLU tasks. In *Proceedings of the Joint 25th Nordic Conference on Computational Linguistics and 11th Baltic Conference on Human Language Technologies (NoDaLiDa/Baltic-HLT 2025)*, pages 561–572, Tallinn, Estonia. University of Tartu Library.

tsroten. 2024. hanzidentifier. https://github.com/tsroten/hanzidentifier. GitHub Repository, version v1.3.0, accessed 2025.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. Multilingual e5 text embeddings: A technical report. *arXiv preprint arXiv:2402.05672*.

Dirk Weissenborn, Georg Wiese, and Laura Seiffe. 2017. Making neural qa as simple as possible but not simpler. *CoNLL 2017*, page 271.

Evan You. 2025. VueJS. https://github.com/vuejs/core. GitHub Repository, version v3.6.0-alpha.2, accessed 2025.

8. Language Resource References

A. Appendix

ab	ban	cr	fat	gur	iu	lad	mni	om	ru	ss	tt	yue
ace	bar	crh	ff	guw	ja	lb	mnw	or	rue	st	tum	za
ady	bcl	cs	fi	gv	jam	lbe	mr	os	rw	stq	tw	zea
af	be	csb	fj	ha	jbo	lez	mrj	pa	sa	su	ty	zh-cn
als	bg	cu	fo	hak	ka	lfn	ms	pag	sah	sv	tyv	zh-tw
alt	bi	cv	fon	haw	kaa	lg	mt	pam	sat	sw	udm	zu
am	bjn	cy	fr	he	kab	li	mwl	pap	sc	szl	ug	
ami	blk	da	frp	hi	kbd	lij	my	pcd	scn	szy	uk	
an	bm	dag	frr	hif	kbp	lld	myv	pcm	sco	ta	ur	
ang	bn	de	fur	hr	kcg	lmo	mzn	pdc	sd	tay	uz	
anp	bo	din	fy	hsb	kg	ln	nap	pfl	se	tcy	ve	
ar	bpy	diq	ga	ht	ki	lo	nds	pi	sg	te	vec	
arc	br	dsb	gag	hu	kk	lt	ne	pl	shi	tet	vep	
ary	bs	dty	gan	hy	kl	ltg	new	pms	shn	tg	vi	
arz	bug	dv	gcr	hyw	km	lv	nia	pnb	si	th	vls	
as	bxr	dz	gd	ia	kn	mad	nl	pnt	sk	ti	vo	
ast	ca	ee	gl	id	ko	mai	nn	ps	skr	tk	wa	
atj	cdo	el	glk	ie	koi	mdf	no	pt-br	sl	tl	war	
av	ce	en	gn	ig	krc	mg	nov	pt-pt	sm	tly	wo	
avk	ceb	eo	gom	ik	ks	mhr	nqo	pwn	smn	tn	wuu	
awa	ch	es	gor	ilo	ku	mi	nso	qu	sn	to	xal	
ay	chr	et	got	inh	kv	min	nv	rm	so	tpi	xh	
az	chy	eu	gpe	io	kw	mk	ny	rmy	sq	tr	xmf	
azb	ckb	ext	gu	is	ky	ml	oc	rn	sr	trv	yi	
ba	co	fa	guc	it	la	mn	olo	ro	srn	ts	yo	

Table 2: List of all languages in MultiWikiQA.

Language	Samples	Language	Samples	Language	Samples	Language	Samples
lld	4,745	tay	3,049	guc	1,558	cu	443
tn	4,744	ami	2,920	pwn	1,471	za	427
pcm	4,623	nia	2,804	mi	1,419	ki	416
gcr	4,590	ny	2,751	awa	1,385	tpi	397
fat	4,539	ff	2,701	pdc	1,381	ti	385
om	4,458	dz	2,679	jam	1,345	got	383
av	4,375	shi	2,677	sm	1,278	ady	380
se	4,257	wo	2,624	st	1,270	lbe	378
tum	4,252	kbd	2,585	ee	1,223	ve	369
gpe	4,242	bpy	2,561	kcg	1,202	srn	321
csb	4,199	ln	2,297	to	1,187	kg	261
mrj	4,194	ace	2,210	tly	1,108	arc	251
gor	4,137	mni	2,179	nov	1,066	chr	185
crh	4,053	mad	2,032	atj	1,064	bi	149
gur	4,042	jbo	2,011	fon	899	iu	148
dty	4,029	hak	1,945	nso	810	ch	135
mdf	3,957	haw	1,903	pag	802	ty	129
xh	3,955	ss	1,856	rn	763	bug	119
krc	3,894	cdo	1,780	fj	745	sg	83
frp	3,849	inh	1,738	rmy	722	pi	79
guw	3,782	din	1,716	gan	697	ik	67
anp	3,667	ltg	1,636	nv	681	cr	33
koi	3,539	ab	1,625	bm	663	chy	25
gag	3,409	tet	1,604	kl	529		
glk	3,384	ts	1,583	pnt	529		
ang	3,376	ks	1,571	xal	464		

Table 3: All the languages in MultiWikiQA with fewer than 5,000 samples.

Language	Mistral Base	Mistral Instruct	Llama Base	Llama Instruct	XLM-RoBERTa	Multi-E5
ace	45.2% ± 2.2%	49.9% ± 4.1%	39.1% ± 8.2%	50.5% ± 2.5%	10.5% ± 5.6%	20.6% ± 13.9%
af	78.0% ± 4.0%	76.5% ± 3.1%	69.7% ± 5.8%	74.5% ± 1.1%	30.9% ± 2.8%	29.1% ± 3.6%
alt	53.1% ± 5.7%	57.0% ± 3.7%	41.6% ± 8.6%	53.8% ± 2.5%	22.7% ± 4.1%	20.1% ± 9.5%
am	21.2% ± 2.6%	22.1% ± 2.2%	18.7% ± 8.6%	20.6% ± 1.5%	21.2% ± 2.0%	20.7% ± 9.1%
ami	14.9% ± 9.1%	20.2% ± 8.0%	16.5% ± 8.1%	25.0% ± 4.7%	0.6% ± 1.4%	0.0% ± 0.0%
an	71.3% ± 2.5%	69.4% ± 2.8%	66.7% ± 3.7%	63.6% ± 2.4%	26.8% ± 2.5%	29.0% ± 2.8%
ang	55.2% ± 3.3%	59.1% ± 3.1%	45.3% ± 2.8%	48.9% ± 3.0%	8.6% ± 6.4%	10.8% ± 10.1%
anp	57.3% ± 4.7%	57.2% ± 4.2%	54.2% ± 3.4%	50.5% ± 3.0%	24.6% ± 4.1%	25.5% ± 4.4%
ar	68.8% ± 6.1%	65.8% ± 4.1%	63.9% ± 5.1%	59.6% ± 1.8%	25.4% ± 3.5%	27.5% ± 4.1%
ary	59.0% ± 4.5%	59.5% ± 2.8%	50.9% ± 5.0%	50.4% ± 3.0%	17.5% ± 3.4%	22.8% ± 5.1%
arz	71.9% ± 6.4%	71.9% ± 5.4%	57.5% ± 5.4%	60.4% ± 5.9%	51.5% ± 2.2%	52.9% ± 2.3%
as	62.1% ± 2.5%	59.2% ± 2.9%	58.2% ± 3.2%	59.5% ± 1.7%	20.5% ± 8.8%	22.8% ± 7.7%
ast	73.3% ± 2.9%	71.9% ± 3.6%	69.5% ± 4.0%	64.4% ± 2.5%	31.6% ± 6.6%	37.1% ± 3.1%
av	36.5% ± 11.3%	45.1% ± 5.0%	31.0% ± 11.8%	40.5% ± 8.6%	8.0% ± 6.5%	7.6% ± 8.4%
avk	39.7% ± 6.6%	44.7% ± 3.4%	10.0% ± 7.8%	46.8% ± 2.9%	33.5% ± 3.0%	37.7% ± 2.7%
awa	67.4% ± 6.3%	68.8% ± 4.9%	63.6% ± 6.7%	67.0% ± 4.5%	31.1% ± 15.8%	35.1% ± 16.0%
ay	33.2% ± 5.3%	37.7% ± 3.7%	13.4% ± 6.4%	36.1% ± 2.8%	11.4% ± 2.7%	24.4% ± 3.5%
az	71.5% ± 8.8%	69.0% ± 4.4%	62.2% ± 7.0%	64.2% ± 3.5%	31.1% ± 1.7%	31.6% ± 3.1%
azb	59.9% ± 5.1%	56.3% ± 10.0%	43.4% ± 7.6%	48.3% ± 6.0%	11.4% ± 7.5%	25.2% ± 5.3%
ba	65.8% ± 6.1%	69.5% ± 4.0%	63.2% ± 3.8%	67.0% ± 2.9%	15.0% ± 6.9%	26.6% ± 5.5%
ban	60.5% ± 2.8%	65.3% ± 2.1%	44.7% ± 11.6%	61.1% ± 3.7%	26.7% ± 3.9%	31.8% ± 3.4%
bar	61.0% ± 4.3%	65.7% ± 3.5%	53.8% ± 4.6%	53.7% ± 3.2%	13.1% ± 6.4%	20.9% ± 3.1%
bcl	64.3% ± 5.4%	70.1% ± 3.5%	54.0% ± 16.0%	65.2% ± 4.6%	18.4% ± 12.5%	27.3% ± 3.3%
be	65.5% ± 6.2%	62.5% ± 3.7%	61.1% ± 3.9%	50.0% ± 2.8%	31.3% ± 3.1%	34.4% ± 2.2%
bg	77.0% ± 4.2%	74.2% ± 3.7%	72.4% ± 4.2%	65.8% ± 2.8%	30.2% ± 2.1%	31.6% ± 3.1%
bjn	68.3% ± 2.2%	68.3% ± 2.6%	66.6% ± 4.2%	69.9% ± 2.3%	27.5% ± 5.3%	33.0% ± 3.5%
blk	16.4% ± 5.3%	21.4% ± 2.6%	13.7% ± 2.5%	14.7% ± 2.5%	1.4% ± 3.1%	0.0% ± 0.0%
bn	62.9% ± 6.2%	64.7% ± 2.9%	55.7% ± 7.7%	62.8% ± 2.2%	27.8% ± 4.9%	29.4% ± 3.0%
bo	0.3% ± 0.4%	0.9% ± 0.8%	3.2% ± 1.6%	3.6% ± 1.8%	0.0% ± 0.1%	0.0% ± 0.0%
bpy	60.9% ± 14.7%	62.8% ± 7.0%	59.2% ± 11.5%	67.6% ± 1.5%	35.3% ± 4.1%	37.0% ± 3.9%
br	68.0% ± 3.6%	67.9% ± 2.0%	58.4% ± 5.2%	63.4% ± 2.2%	27.0% ± 3.4%	28.5% ± 3.1%
bs	69.6% ± 5.4%	69.0% ± 2.9%	61.6% ± 5.2%	66.2% ± 2.2%	36.0% ± 2.6%	36.0% ± 3.2%
bxr	56.4% ± 4.6%	55.9% ± 7.1%	25.4% ± 13.1%	49.1% ± 4.5%	19.8% ± 2.7%	21.4% ± 9.5%
ca	73.9% ± 4.6%	71.3% ± 4.3%	69.8% ± 3.9%	66.8% ± 2.7%	30.6% ± 4.5%	30.5% ± 3.6%
cdo	34.4% ± 6.4%	41.0% ± 3.5%	45.2% ± 6.6%	42.3% ± 10.2%	6.2% ± 8.1%	0.8% ± 3.1%
ce	49.4% ± 6.7%	55.1% ± 4.1%	17.7% ± 6.6%	45.2% ± 3.5%	19.4% ± 3.7%	24.7% ± 4.9%
ceb	69.2% ± 7.7%	70.7% ± 5.5%	43.9% ± 6.6%	61.0% ± 7.0%	44.5% ± 2.8%	44.9% ± 2.6%
ckb	35.8% ± 9.1%	27.9% ± 7.9%	46.3% ± 4.8%	48.4% ± 3.4%	1.1% ± 2.3%	18.4% ± 8.4%
co	69.6% ± 6.9%	64.3% ± 4.3%	64.3% ± 8.3%	59.3% ± 5.6%	26.0% ± 11.7%	33.5% ± 4.5%
crh	63.5% ± 4.9%	63.2% ± 3.9%	54.3% ± 4.2%	59.2% ± 3.1%	21.4% ± 3.2%	28.7% ± 3.6%
cs	71.3% ± 3.7%	68.2% ± 4.1%	68.9% ± 3.3%	64.0% ± 3.1%	29.8% ± 2.4%	31.2% ± 3.6%
csb	53.9% ± 7.0%	55.2% ± 3.9%	45.4% ± 6.3%	45.1% ± 5.5%	14.3% ± 7.0%	21.1% ± 3.0%
cv	37.4% ± 5.7%	41.8% ± 3.2%	23.0% ± 10.9%	58.3% ± 3.5%	11.6% ± 2.9%	19.5% ± 5.3%
cy	90.3% ± 1.9%	90.6% ± 1.3%	67.1% ± 13.0%	89.2% ± 1.9%	49.4% ± 2.6%	49.4% ± 2.2%
da	78.9% ± 2.7%	77.1% ± 2.3%	75.2% ± 2.3%	74.2% ± 2.0%	32.3% ± 3.9%	33.1% ± 2.4%
dag	29.4% ± 11.3%	36.4% ± 6.0%	22.1% ± 11.5%	43.8% ± 1.6%	8.2% ± 4.5%	6.4% ± 9.4%
de	80.3% ± 2.2%	76.6% ± 2.4%	79.2% ± 2.1%	70.1% ± 2.4%	28.5% ± 3.1%	30.5% ± 3.6%
din	25.9% ± 8.6%	27.4% ± 7.1%	14.2% ± 6.2%	23.3% ± 5.7%	7.0% ± 11.3%	7.9% ± 12.6%
diq	43.4% ± 4.1%	48.2% ± 2.9%	31.8% ± 11.1%	41.5% ± 4.7%	13.7% ± 6.6%	14.8% ± 8.8%
dsb	61.7% ± 3.2%	61.4% ± 2.1%	37.9% ± 18.0%	55.4% ± 2.9%	24.1% ± 3.9%	31.0% ± 4.1%
dty	65.9% ± 4.0%	63.1% ± 4.4%	63.8% ± 2.5%	62.4% ± 3.1%	24.6% ± 2.0%	27.1% ± 1.6%
dv	7.2% ± 6.0%	8.5% ± 6.1%	18.7% ± 8.6%	26.3% ± 4.2%	0.1% ± 0.2%	0.2% ± 0.8%
dz	1.4% ± 1.2%	1.9% ± 1.1%	5.6% ± 1.5%	5.9% ± 1.4%	0.0% ± 0.0%	0.0% ± 0.0%
el	71.6% ± 4.1%	69.2% ± 2.4%	68.5% ± 4.3%	58.0% ± 3.3%	27.3% ± 3.2%	29.3% ± 3.3%
en	79.5% ± 3.9%	78.5% ± 2.7%	76.4% ± 4.4%	71.4% ± 1.7%	28.1% ± 3.6%	30.6% ± 1.9%
eo	73.6% ± 5.2%	70.6% ± 4.7%	67.9% ± 5.0%	68.5% ± 2.2%	30.6% ± 3.5%	30.6% ± 3.7%
es	76.6% ± 2.6%	74.1% ± 2.1%	72.5% ± 4.7%	67.9% ± 2.7%	29.4% ± 2.6%	27.7% ± 4.4%
et	73.5% ± 3.6%	71.8% ± 4.0%	67.8% ± 5.0%	65.6% ± 2.1%	30.7% ± 2.3%	29.1% ± 2.4%
eu	68.8% ± 2.9%	63.1% ± 5.2%	62.4% ± 4.1%	63.8% ± 3.4%	27.7% ± 3.9%	26.9% ± 3.3%
ext	68.2% ± 4.4%	70.0% ± 4.1%	65.1% ± 4.5%	58.6% ± 4.0%	21.7% ± 4.1%	27.2% ± 2.4%
fa	68.5% ± 3.0%	66.8% ± 1.6%	62.0% ± 3.6%	62.9% ± 2.0%	32.4% ± 3.4%	32.8% ± 2.5%
fat	30.3% ± 5.0%	36.6% ± 3.5%	24.1% ± 7.7%	38.0% ± 3.5%	1.0% ± 2.4%	8.4% ± 9.2%
ff	20.4% ± 8.8%	26.0% ± 5.2%	14.9% ± 4.6%	31.6% ± 2.9%	6.3% ± 5.6%	4.5% ± 8.8%
fi	69.4% ± 5.6%	69.8% ± 5.1%	62.3% ± 4.5%	63.7% ± 1.9%	29.7% ± 3.4%	28.2% ± 4.2%
fo	65.7% ± 2.0%	67.7% ± 2.9%	63.8% ± 4.1%	66.4% ± 3.3%	24.4% ± 4.8%	27.6% ± 4.8%

Table 4: The F1-scores on MultiWikiQA for languages 1-66, sorted alphabetically.

Language	Mistral Base	Mistral Instruct	Llama Base	Llama Instruct	XLM-RoBERTa	Multi-E5
fr	74.1% ± 4.7%	71.6% ± 1.6%	71.5% ± 3.7%	66.5% ± 1.4%	28.0% ± 2.4%	29.1% ± 4.7%
frp	62.5% ± 3.7%	59.2% ± 3.8%	54.8% ± 2.8%	45.1% ± 3.3%	14.4% ± 6.4%	26.2% ± 3.5%
frr	53.9% ± 4.2%	58.7% ± 2.9%	49.1% ± 5.1%	53.5% ± 2.5%	17.5% ± 2.5%	23.5% ± 3.0%
fur	63.0% ± 5.6%	64.8% ± 4.6%	54.0% ± 9.0%	60.7% ± 2.5%	7.8% ± 4.2%	25.8% ± 2.6%
fy	68.0% ± 4.0%	69.5% ± 2.4%	68.6% ± 6.2%	73.2% ± 2.1%	27.2% ± 3.1%	29.1% ± 3.3%
ga	68.5% ± 6.0%	69.7% ± 3.3%	56.6% ± 7.7%	62.9% ± 4.9%	24.2% ± 2.7%	26.9% ± 4.3%
gag	65.2% ± 2.5%	66.1% ± 1.4%	57.1% ± 6.4%	57.8% ± 1.7%	27.3% ± 4.5%	33.5% ± 3.4%
gcr	70.7% ± 5.4%	72.8% ± 3.2%	61.0% ± 4.1%	59.2% ± 6.7%	26.7% ± 2.3%	34.8% ± 2.3%
gd	70.3% ± 3.6%	66.8% ± 3.2%	58.0% ± 7.2%	64.7% ± 1.7%	31.2% ± 3.1%	30.4% ± 2.5%
gl	78.4% ± 3.9%	75.4% ± 4.7%	71.9% ± 4.5%	66.4% ± 2.5%	31.2% ± 2.5%	29.6% ± 5.1%
glk	39.9% ± 9.0%	37.6% ± 6.5%	33.9% ± 5.5%	33.8% ± 3.4%	13.2% ± 4.2%	14.2% ± 12.7%
gn	21.4% ± 6.2%	26.6% ± 4.2%	22.3% ± 9.4%	37.0% ± 2.3%	1.8% ± 2.2%	1.7% ± 6.6%
gom	42.5% ± 2.8%	47.1% ± 2.6%	41.3% ± 5.3%	47.7% ± 2.5%	8.4% ± 5.1%	6.0% ± 8.6%
gor	61.4% ± 3.9%	64.0% ± 2.8%	54.1% ± 5.5%	60.9% ± 3.5%	31.3% ± 2.2%	33.8% ± 2.1%
gpe	75.0% ± 2.5%	74.5% ± 1.9%	71.0% ± 3.8%	69.3% ± 2.5%	32.1% ± 2.9%	33.5% ± 3.7%
gu	61.2% ± 4.2%	62.3% ± 3.8%	57.4% ± 1.6%	53.9% ± 3.4%	28.5% ± 3.1%	27.2% ± 5.0%
guc	19.6% ± 6.2%	22.7% ± 5.0%	8.7% ± 3.6%	18.5% ± 3.1%	0.6% ± 2.3%	0.0% ± 0.1%
gur	37.2% ± 5.7%	39.7% ± 3.6%	20.9% ± 9.6%	38.5% ± 1.8%	10.1% ± 6.1%	16.4% ± 7.0%
guw	17.5% ± 9.2%	24.8% ± 4.8%	24.1% ± 5.0%	32.8% ± 2.5%	0.2% ± 0.7%	5.0% ± 8.0%
gv	52.2% ± 4.5%	53.8% ± 2.5%	43.0% ± 8.6%	51.8% ± 2.0%	13.8% ± 3.3%	15.4% ± 7.5%
ha	52.4% ± 4.9%	55.6% ± 2.6%	57.6% ± 3.4%	58.3% ± 2.3%	25.4% ± 4.1%	25.9% ± 5.2%
hak	37.4% ± 8.6%	39.5% ± 7.5%	40.8% ± 10.6%	44.2% ± 6.8%	5.9% ± 7.5%	6.2% ± 9.8%
haw	38.5% ± 11.3%	46.8% ± 7.8%	24.0% ± 8.7%	40.9% ± 6.4%	4.5% ± 6.7%	8.4% ± 12.7%
he	73.2% ± 3.8%	69.2% ± 3.1%	63.7% ± 3.9%	63.6% ± 2.8%	24.2% ± 3.4%	26.2% ± 4.1%
hi	67.6% ± 5.3%	67.4% ± 3.5%	65.5% ± 8.0%	65.9% ± 2.1%	28.5% ± 1.5%	28.5% ± 2.6%
hif	70.6% ± 4.3%	67.2% ± 2.0%	60.8% ± 3.9%	66.3% ± 2.8%	25.0% ± 3.5%	26.3% ± 3.6%
hr	64.6% ± 5.0%	64.7% ± 2.5%	64.3% ± 2.8%	60.9% ± 2.4%	28.1% ± 2.7%	30.2% ± 2.6%
hsb	60.2% ± 4.1%	63.9% ± 2.4%	47.0% ± 11.7%	52.8% ± 3.5%	28.5% ± 4.7%	33.9% ± 3.9%
ht	79.3% ± 4.0%	76.9% ± 3.2%	67.1% ± 2.9%	70.5% ± 2.3%	36.3% ± 1.8%	44.2% ± 1.4%
hu	66.6% ± 7.5%	65.3% ± 5.9%	63.2% ± 5.7%	63.2% ± 3.8%	25.8% ± 3.0%	28.4% ± 1.7%
hy	64.0% ± 10.4%	62.6% ± 6.9%	47.6% ± 5.1%	48.9% ± 3.2%	32.6% ± 2.0%	33.1% ± 2.6%
hyw	56.3% ± 7.9%	54.0% ± 8.0%	44.5% ± 2.6%	44.8% ± 3.0%	29.0% ± 2.8%	29.8% ± 2.0%
ia	72.1% ± 2.9%	69.1% ± 3.4%	63.8% ± 2.3%	60.4% ± 2.7%	28.7% ± 2.3%	33.6% ± 2.5%
id	77.9% ± 3.9%	75.0% ± 3.5%	75.0% ± 3.4%	71.0% ± 2.5%	34.9% ± 3.0%	35.7% ± 3.2%
ie	76.3% ± 4.3%	75.6% ± 4.4%	73.4% ± 3.3%	72.7% ± 2.5%	35.1% ± 1.8%	36.0% ± 2.8%
ig	45.2% ± 5.6%	53.3% ± 3.4%	49.0% ± 6.9%	60.7% ± 2.1%	2.5% ± 2.7%	18.4% ± 12.5%
ilo	56.2% ± 4.8%	62.2% ± 2.4%	37.8% ± 16.2%	61.4% ± 2.6%	9.0% ± 6.0%	24.0% ± 5.9%
inh	33.7% ± 6.9%	39.2% ± 6.1%	18.3% ± 7.3%	29.3% ± 4.3%	13.6% ± 8.6%	11.2% ± 14.5%
io	68.5% ± 5.2%	64.2% ± 5.0%	68.1% ± 3.8%	66.7% ± 2.3%	39.3% ± 3.4%	42.2% ± 1.8%
is	70.7% ± 2.8%	68.6% ± 2.5%	66.0% ± 4.1%	63.5% ± 1.6%	27.7% ± 2.0%	29.5% ± 3.5%
it	76.3% ± 3.5%	73.0% ± 1.9%	72.3% ± 4.4%	65.0% ± 1.5%	25.0% ± 3.2%	25.8% ± 3.1%
ja	50.0% ± 3.4%	47.4% ± 4.3%	48.1% ± 4.3%	41.5% ± 3.5%	14.9% ± 3.6%	17.8% ± 3.6%
jam	54.5% ± 6.2%	54.6% ± 5.4%	39.1% ± 4.9%	38.5% ± 6.8%	3.9% ± 6.4%	19.7% ± 6.7%
ka	66.4% ± 7.0%	62.4% ± 3.9%	51.7% ± 2.2%	52.7% ± 1.9%	32.7% ± 3.8%	31.9% ± 2.1%
kaa	58.1% ± 5.9%	57.1% ± 4.6%	46.2% ± 5.2%	48.5% ± 3.1%	13.9% ± 4.9%	17.1% ± 8.7%
kab	22.7% ± 7.3%	28.5% ± 6.3%	14.9% ± 4.4%	30.3% ± 2.4%	1.5% ± 2.3%	3.5% ± 6.4%
kbd	32.4% ± 5.8%	36.9% ± 5.0%	27.7% ± 7.3%	29.1% ± 3.6%	5.3% ± 7.5%	0.0% ± 0.0%
kbp	13.4% ± 11.7%	18.9% ± 13.3%	13.0% ± 7.1%	30.8% ± 3.7%	3.0% ± 4.6%	6.6% ± 12.0%
kk	69.6% ± 8.1%	63.6% ± 9.0%	60.6% ± 7.0%	59.4% ± 5.5%	32.7% ± 3.0%	32.4% ± 5.5%
km	7.7% ± 5.0%	7.4% ± 3.8%	26.4% ± 8.3%	24.6% ± 3.4%	25.3% ± 3.4%	27.7% ± 3.6%
kn	66.7% ± 2.8%	63.4% ± 1.9%	53.8% ± 3.3%	54.6% ± 2.1%	29.2% ± 6.6%	26.6% ± 8.5%
ko	60.8% ± 5.5%	57.0% ± 4.3%	56.4% ± 5.1%	51.8% ± 2.7%	18.0% ± 12.1%	17.6% ± 10.5%
koi	37.1% ± 9.8%	47.5% ± 3.0%	7.5% ± 6.3%	45.8% ± 2.2%	17.9% ± 8.8%	17.6% ± 13.1%
krc	59.9% ± 3.5%	58.4% ± 3.9%	48.7% ± 3.3%	52.2% ± 2.0%	23.1% ± 8.5%	26.9% ± 11.6%
ks	44.8% ± 4.4%	45.8% ± 3.5%	38.7% ± 5.6%	42.2% ± 5.9%	11.7% ± 7.9%	25.2% ± 11.7%
ku	62.5% ± 5.0%	67.2% ± 2.6%	46.2% ± 5.3%	51.2% ± 2.8%	21.5% ± 8.5%	22.7% ± 4.7%
kv	48.5% ± 6.7%	50.7% ± 6.9%	24.9% ± 14.0%	50.6% ± 3.4%	19.5% ± 5.7%	24.0% ± 5.2%
kw	41.7% ± 11.3%	47.5% ± 5.1%	24.9% ± 12.1%	51.8% ± 3.5%	13.0% ± 4.0%	15.5% ± 4.8%
ky	59.8% ± 9.9%	58.2% ± 6.0%	56.7% ± 2.9%	60.0% ± 3.2%	31.6% ± 2.1%	30.2% ± 2.2%
la	67.5% ± 5.8%	67.3% ± 4.7%	58.6% ± 4.4%	63.1% ± 2.6%	25.5% ± 2.3%	28.6% ± 3.7%
lad	70.2% ± 2.6%	66.8% ± 2.8%	63.5% ± 3.0%	55.1% ± 2.8%	26.8% ± 3.5%	32.1% ± 3.2%
lb	71.0% ± 3.8%	73.0% ± 6.3%	65.4% ± 4.0%	68.3% ± 2.6%	16.7% ± 4.5%	31.3% ± 4.1%
lez	48.0% ± 5.0%	52.4% ± 4.0%	42.1% ± 4.2%	49.3% ± 2.5%	15.7% ± 6.4%	19.4% ± 7.1%
lfn	71.0% ± 2.4%	67.1% ± 1.8%	62.4% ± 3.7%	59.8% ± 2.2%	23.9% ± 4.2%	27.3% ± 4.9%
lg	28.4% ± 11.8%	42.4% ± 6.4%	20.0% ± 7.1%	39.7% ± 1.6%	3.6% ± 4.7%	14.3% ± 9.0%

Table 5: The F1-scores on MultiWikiQA for languages 66-131, sorted alphabetically.

Language	Mistral Base	Mistral Instruct	Llama Base	Llama Instruct	XLM-RoBERTa	Multi-E5
li	64.6% ± 3.3%	66.2% ± 2.9%	60.9% ± 2.9%	62.3% ± 1.8%	18.2% ± 8.7%	24.3% ± 4.5%
lij	66.9% ± 2.6%	64.4% ± 3.9%	58.2% ± 3.1%	55.1% ± 1.8%	9.6% ± 7.5%	19.7% ± 6.4%
lld	51.9% ± 9.8%	54.0% ± 1.9%	39.3% ± 14.9%	51.7% ± 3.3%	8.8% ± 6.8%	21.8% ± 9.5%
lmo	64.9% ± 4.2%	66.0% ± 4.7%	54.6% ± 11.3%	57.9% ± 2.3%	11.4% ± 6.2%	22.4% ± 4.4%
ln	32.9% ± 19.6%	44.8% ± 12.0%	26.8% ± 12.0%	46.3% ± 2.9%	4.3% ± 3.6%	25.5% ± 7.2%
lo	13.0% ± 5.7%	11.9% ± 3.9%	9.8% ± 7.7%	21.5% ± 2.5%	23.5% ± 4.0%	24.8% ± 8.1%
lt	70.1% ± 5.2%	67.9% ± 4.7%	68.1% ± 3.0%	64.3% ± 2.6%	33.9% ± 2.1%	34.6% ± 4.4%
ltg	61.1% ± 5.4%	64.4% ± 3.9%	38.7% ± 6.8%	52.2% ± 6.6%	19.0% ± 13.0%	24.9% ± 13.2%
lv	67.3% ± 5.9%	70.0% ± 4.1%	60.6% ± 5.7%	65.7% ± 1.6%	33.1% ± 2.5%	30.3% ± 13.0%
mad	56.8% ± 5.4%	59.5% ± 3.5%	58.1% ± 4.6%	59.4% ± 2.9%	11.9% ± 8.8%	23.4% ± 5.6%
mai	63.3% ± 5.8%	64.0% ± 4.4%	59.3% ± 3.7%	60.2% ± 3.0%	29.4% ± 2.6%	29.4% ± 5.1%
mdf	36.9% ± 10.0%	46.1% ± 3.1%	14.6% ± 6.8%	42.6% ± 2.3%	22.6% ± 2.7%	22.7% ± 3.0%
mg	48.8% ± 7.1%	54.3% ± 4.0%	21.5% ± 9.5%	45.0% ± 2.5%	34.7% ± 14.0%	35.0% ± 16.3%
mhr	48.9% ± 5.7%	53.2% ± 3.6%	31.3% ± 10.5%	53.9% ± 3.6%	25.3% ± 6.1%	30.1% ± 4.9%
mi	64.8% ± 3.9%	66.4% ± 3.9%	54.3% ± 11.4%	58.4% ± 5.3%	16.7% ± 11.0%	34.9% ± 6.2%
min	66.5% ± 3.3%	67.1% ± 3.0%	64.7% ± 4.1%	62.0% ± 2.1%	26.2% ± 2.8%	29.1% ± 3.0%
mk	74.4% ± 3.9%	72.3% ± 2.1%	70.6% ± 3.7%	73.2% ± 1.8%	31.3% ± 3.1%	31.5% ± 2.1%
ml	64.9% ± 1.5%	64.9% ± 2.8%	52.6% ± 1.9%	52.5% ± 2.4%	29.9% ± 3.5%	30.8% ± 2.6%
mn	63.3% ± 6.0%	64.0% ± 4.5%	52.6% ± 11.2%	57.2% ± 2.4%	28.9% ± 2.4%	29.1% ± 2.3%
mni	1.1% ± 0.7%	1.0% ± 0.6%	2.0% ± 0.7%	2.7% ± 1.5%	0.0% ± 0.0%	0.0% ± 0.0%
mnw	4.7% ± 4.1%	7.7% ± 3.5%	12.8% ± 4.5%	21.2% ± 4.3%	0.1% ± 0.3%	0.0% ± 0.0%
mr	71.8% ± 3.8%	71.8% ± 1.8%	65.2% ± 6.1%	72.8% ± 1.8%	33.3% ± 3.3%	32.2% ± 5.8%
mrj	44.0% ± 9.7%	49.9% ± 5.7%	37.8% ± 10.1%	52.8% ± 2.6%	17.5% ± 3.3%	23.6% ± 9.2%
ms	81.6% ± 2.2%	77.1% ± 2.2%	76.7% ± 1.6%	76.1% ± 2.3%	33.3% ± 3.5%	32.3% ± 3.1%
mt	61.7% ± 4.7%	61.5% ± 3.6%	56.1% ± 3.6%	59.9% ± 1.7%	1.9% ± 2.8%	18.2% ± 9.3%
mwl	76.1% ± 4.3%	73.9% ± 1.9%	67.4% ± 7.6%	60.6% ± 2.4%	26.3% ± 4.4%	31.3% ± 5.8%
my	26.4% ± 7.8%	29.8% ± 5.8%	19.2% ± 5.9%	23.0% ± 5.1%	24.2% ± 5.6%	23.0% ± 4.4%
myv	39.7% ± 4.0%	43.2% ± 3.2%	27.4% ± 8.3%	43.2% ± 2.5%	13.4% ± 5.9%	16.2% ± 10.7%
mzn	56.3% ± 7.2%	57.5% ± 5.4%	50.9% ± 6.3%	51.0% ± 3.5%	19.7% ± 3.2%	23.8% ± 2.1%
nap	66.8% ± 3.4%	67.1% ± 3.3%	57.1% ± 4.2%	58.3% ± 2.1%	8.9% ± 8.9%	24.9% ± 5.0%
nds	72.5% ± 3.1%	71.5% ± 3.4%	66.6% ± 4.2%	69.9% ± 2.4%	25.8% ± 5.4%	31.7% ± 4.4%
ne	62.8% ± 4.7%	62.4% ± 3.5%	65.0% ± 2.8%	65.4% ± 2.4%	29.5% ± 1.3%	29.3% ± 3.0%
new	39.4% ± 15.3%	46.1% ± 12.1%	23.2% ± 16.7%	50.4% ± 10.0%	46.1% ± 3.9%	40.7% ± 17.5%
nia	24.2% ± 6.9%	27.1% ± 5.1%	15.9% ± 5.8%	28.4% ± 2.3%	0.5% ± 1.5%	0.2% ± 0.9%
nl	78.3% ± 4.0%	78.7% ± 2.8%	74.6% ± 4.2%	69.0% ± 1.7%	30.4% ± 2.8%	31.1% ± 3.3%
nn	77.9% ± 4.5%	76.2% ± 2.8%	74.6% ± 4.7%	69.2% ± 1.7%	32.4% ± 3.2%	32.7% ± 3.4%
no	77.1% ± 4.3%	76.7% ± 2.3%	74.2% ± 3.7%	70.4% ± 2.7%	31.6% ± 2.8%	33.4% ± 2.2%
nqo	2.5% ± 0.9%	2.3% ± 0.9%	2.6% ± 0.4%	2.5% ± 0.6%	0.0% ± 0.1%	0.0% ± 0.1%
ny	51.5% ± 4.1%	56.1% ± 3.6%	33.6% ± 7.4%	45.6% ± 2.4%	7.2% ± 5.5%	24.1% ± 4.7%
oc	65.0% ± 4.8%	63.6% ± 2.2%	59.2% ± 4.0%	58.7% ± 1.3%	23.3% ± 2.6%	29.5% ± 1.8%
olo	60.0% ± 4.4%	61.9% ± 3.1%	53.8% ± 4.0%	55.1% ± 3.0%	24.0% ± 2.9%	27.4% ± 3.6%
om	18.2% ± 8.2%	27.3% ± 4.9%	13.0% ± 6.6%	21.7% ± 9.1%	9.6% ± 5.5%	9.9% ± 9.3%
or	33.2% ± 1.8%	31.1% ± 2.1%	40.8% ± 5.4%	41.3% ± 1.6%	28.8% ± 3.2%	32.3% ± 3.7%
os	44.3% ± 6.2%	48.6% ± 4.3%	41.4% ± 6.4%	54.1% ± 3.6%	14.1% ± 4.2%	11.8% ± 13.3%
pa	73.9% ± 3.4%	74.0% ± 2.6%	60.0% ± 4.6%	55.2% ± 5.2%	27.4% ± 4.2%	31.1% ± 2.9%
pam	40.1% ± 15.2%	52.3% ± 2.4%	39.8% ± 9.7%	54.8% ± 2.3%	16.5% ± 3.2%	25.1% ± 2.2%
pap	74.0% ± 2.0%	73.9% ± 2.4%	68.7% ± 4.9%	67.2% ± 3.7%	25.4% ± 3.3%	30.5% ± 2.6%
pcd	70.6% ± 5.3%	68.5% ± 4.3%	63.7% ± 4.9%	60.9% ± 3.0%	20.7% ± 8.5%	31.3% ± 2.2%
pcm	78.3% ± 1.7%	76.6% ± 2.0%	73.6% ± 2.2%	66.5% ± 1.9%	22.4% ± 9.9%	30.9% ± 2.1%
pdc	58.2% ± 7.8%	66.7% ± 7.0%	51.7% ± 5.0%	47.8% ± 4.5%	15.1% ± 9.6%	25.0% ± 9.3%
pfl	64.3% ± 2.0%	68.4% ± 1.7%	55.6% ± 4.3%	52.8% ± 4.8%	17.8% ± 4.2%	21.3% ± 5.4%
pl	69.6% ± 5.5%	64.7% ± 3.8%	58.1% ± 6.7%	55.8% ± 2.6%	28.2% ± 5.5%	32.8% ± 4.3%
pms	66.7% ± 3.4%	67.2% ± 4.0%	59.1% ± 5.1%	58.1% ± 2.9%	15.0% ± 5.6%	21.7% ± 7.2%
pnb	61.6% ± 3.4%	58.9% ± 1.8%	60.1% ± 2.4%	59.7% ± 2.1%	19.1% ± 5.3%	25.3% ± 9.0%
ps	49.9% ± 6.7%	50.9% ± 4.2%	44.7% ± 2.9%	41.8% ± 4.1%	26.5% ± 5.2%	24.8% ± 5.8%
pt-br	79.2% ± 5.3%	75.0% ± 6.2%	76.3% ± 3.9%	69.3% ± 2.6%	28.7% ± 3.2%	29.3% ± 4.4%
pt-pt	77.7% ± 1.5%	74.1% ± 2.3%	74.2% ± 3.9%	67.0% ± 1.6%	31.0% ± 3.5%	30.0% ± 6.3%
pwn	20.4% ± 7.4%	24.2% ± 5.5%	15.2% ± 5.5%	25.4% ± 4.7%	3.1% ± 5.2%	5.8% ± 9.2%
qu	50.5% ± 12.1%	56.1% ± 3.2%	13.4% ± 7.9%	53.6% ± 2.3%	16.0% ± 3.6%	26.7% ± 4.6%
rm	60.5% ± 4.0%	65.4% ± 3.1%	54.2% ± 2.5%	58.6% ± 2.8%	17.4% ± 2.1%	28.5% ± 3.4%
ro	75.2% ± 2.2%	72.9% ± 2.6%	71.1% ± 3.4%	68.8% ± 2.7%	26.6% ± 2.8%	27.4% ± 5.0%
ru	63.0% ± 6.1%	59.9% ± 5.2%	58.2% ± 5.2%	51.3% ± 4.3%	27.3% ± 1.8%	27.8% ± 2.3%
rue	64.5% ± 2.4%	62.4% ± 2.5%	57.1% ± 3.6%	45.8% ± 2.3%	29.9% ± 3.7%	33.8% ± 3.6%
rw	44.5% ± 3.4%	50.7% ± 3.2%	21.1% ± 8.0%	43.6% ± 1.8%	7.0% ± 3.6%	23.9% ± 4.7%
sa	50.9% ± 4.8%	55.8% ± 4.5%	42.1% ± 13.9%	48.5% ± 2.6%	22.7% ± 1.9%	20.3% ± 9.1%

Table 6: The F1-scores on MultiWikiQA for languages 131-196, sorted alphabetically.

Language	Mistral Base	Mistral Instruct	Llama Base	Llama Instruct	XLM-RoBERTa	Multi-E5
sah	41.1% ± 4.4%	44.8% ± 3.7%	43.6% ± 5.8%	52.0% ± 1.9%	5.8% ± 6.2%	7.9% ± 12.2%
sat	5.0% ± 3.1%	4.7% ± 3.3%	37.3% ± 4.1%	36.2% ± 1.6%	0.0% ± 0.0%	0.0% ± 0.0%
sc	64.0% ± 3.7%	62.8% ± 3.2%	56.8% ± 5.8%	54.3% ± 2.5%	13.7% ± 6.2%	25.1% ± 4.4%
scn	65.4% ± 6.2%	69.3% ± 3.7%	54.2% ± 5.7%	59.2% ± 3.4%	9.2% ± 8.2%	23.7% ± 4.7%
sco	75.6% ± 3.6%	75.7% ± 2.9%	71.4% ± 5.1%	68.4% ± 3.1%	24.7% ± 2.9%	30.7% ± 4.1%
sd	45.7% ± 5.2%	45.1% ± 5.6%	52.6% ± 7.3%	57.6% ± 4.5%	29.3% ± 4.4%	28.1% ± 1.6%
se	54.5% ± 6.6%	58.9% ± 5.0%	29.4% ± 10.3%	46.5% ± 5.2%	13.7% ± 2.7%	16.5% ± 5.1%
shi	24.8% ± 4.3%	27.7% ± 2.3%	22.7% ± 4.6%	23.2% ± 2.8%	0.1% ± 0.2%	0.2% ± 0.6%
shn	3.2% ± 3.6%	5.7% ± 3.7%	16.4% ± 2.9%	14.2% ± 3.1%	0.2% ± 0.4%	0.0% ± 0.0%
si	11.0% ± 7.7%	11.9% ± 3.7%	37.9% ± 4.9%	44.9% ± 2.3%	24.7% ± 2.1%	23.8% ± 2.5%
sk	72.4% ± 4.1%	70.1% ± 3.2%	65.3% ± 4.8%	59.5% ± 1.8%	27.9% ± 4.6%	28.9% ± 3.5%
skr	49.3% ± 2.8%	46.3% ± 4.1%	47.5% ± 2.6%	46.9% ± 2.3%	13.9% ± 6.8%	15.0% ± 7.6%
sl	71.5% ± 7.4%	69.8% ± 4.7%	64.3% ± 5.3%	64.3% ± 2.7%	26.3% ± 5.1%	26.2% ± 6.4%
smn	53.7% ± 2.7%	56.0% ± 2.7%	21.7% ± 13.4%	53.1% ± 2.8%	20.1% ± 5.2%	27.5% ± 4.1%
sn	43.1% ± 5.9%	49.9% ± 5.0%	12.7% ± 6.1%	36.1% ± 4.9%	12.5% ± 11.1%	21.5% ± 10.0%
so	39.1% ± 11.1%	47.7% ± 3.0%	29.5% ± 3.7%	42.7% ± 3.7%	21.8% ± 2.3%	22.1% ± 2.0%
sq	67.8% ± 5.5%	64.3% ± 3.9%	60.6% ± 5.0%	60.3% ± 1.4%	26.5% ± 2.0%	26.1% ± 5.8%
sr	62.3% ± 6.6%	60.6% ± 4.3%	57.9% ± 6.2%	59.9% ± 5.3%	28.7% ± 2.2%	26.4% ± 11.6%
ss	36.6% ± 6.8%	43.6% ± 6.3%	14.4% ± 7.8%	30.8% ± 3.8%	5.2% ± 5.5%	19.7% ± 5.3%
stq	56.4% ± 4.3%	61.6% ± 2.8%	53.8% ± 5.0%	59.0% ± 2.0%	17.0% ± 4.6%	24.2% ± 3.3%
su	65.0% ± 4.1%	69.7% ± 1.7%	43.3% ± 13.6%	70.0% ± 2.9%	25.7% ± 3.5%	28.0% ± 2.5%
sv	78.3% ± 2.4%	79.0% ± 1.8%	72.8% ± 5.1%	70.8% ± 1.6%	33.9% ± 2.8%	33.1% ± 2.9%
sw	77.8% ± 3.4%	79.4% ± 1.1%	69.0% ± 2.2%	72.8% ± 1.4%	30.6% ± 2.7%	30.0% ± 4.2%
szl	58.3% ± 9.0%	59.7% ± 6.8%	50.2% ± 9.9%	47.4% ± 7.2%	16.3% ± 7.6%	26.0% ± 3.1%
szy	15.8% ± 6.1%	24.0% ± 3.2%	11.4% ± 5.2%	29.4% ± 3.3%	7.4% ± 2.4%	2.4% ± 5.6%
ta	61.4% ± 6.3%	65.8% ± 4.2%	56.0% ± 5.3%	59.4% ± 1.9%	34.6% ± 2.9%	31.9% ± 3.1%
tay	35.3% ± 5.4%	39.1% ± 4.6%	10.3% ± 5.6%	35.5% ± 3.1%	16.7% ± 4.9%	18.6% ± 7.1%
tcy	41.0% ± 2.6%	43.5% ± 2.8%	36.2% ± 5.0%	37.4% ± 3.1%	7.4% ± 6.2%	10.6% ± 9.5%
te	74.5% ± 1.4%	71.1% ± 2.0%	60.7% ± 3.2%	60.0% ± 2.2%	31.5% ± 1.7%	27.9% ± 12.7%
tet	66.3% ± 4.5%	67.1% ± 2.5%	53.1% ± 4.9%	62.2% ± 3.5%	8.0% ± 6.6%	21.0% ± 18.2%
tg	63.3% ± 3.1%	66.4% ± 2.1%	61.3% ± 3.8%	60.9% ± 2.2%	3.0% ± 2.3%	15.6% ± 12.3%
th	63.7% ± 2.6%	62.3% ± 2.2%	62.1% ± 3.4%	56.5% ± 3.7%	24.6% ± 11.3%	28.7% ± 2.0%
tk	59.4% ± 2.8%	57.5% ± 3.1%	52.5% ± 3.2%	56.1% ± 1.8%	13.6% ± 3.4%	7.5% ± 10.1%
tl	73.4% ± 3.4%	70.0% ± 2.6%	67.1% ± 4.7%	64.3% ± 2.7%	28.6% ± 2.9%	30.1% ± 2.7%
tn	29.1% ± 12.9%	33.8% ± 14.9%	18.5% ± 15.6%	43.7% ± 5.0%	6.9% ± 5.7%	23.6% ± 4.5%
tr	74.1% ± 3.0%	64.3% ± 2.0%	69.0% ± 5.5%	69.4% ± 1.3%	29.3% ± 4.0%	28.7% ± 4.6%
trv	35.1% ± 2.7%	37.7% ± 2.5%	19.6% ± 6.4%	30.8% ± 5.2%	14.6% ± 7.1%	22.3% ± 4.8%
ts	29.7% ± 9.2%	30.8% ± 8.9%	13.7% ± 5.9%	34.8% ± 7.6%	10.4% ± 10.9%	19.1% ± 20.5%
tt	67.4% ± 21.8%	64.1% ± 20.5%	49.6% ± 16.6%	60.0% ± 3.8%	39.3% ± 2.8%	43.7% ± 4.6%
tum	40.4% ± 11.5%	53.3% ± 7.2%	18.3% ± 8.7%	40.5% ± 7.2%	16.5% ± 4.0%	27.7% ± 12.0%
tw	35.8% ± 3.2%	39.4% ± 2.4%	31.2% ± 8.9%	45.0% ± 2.9%	1.7% ± 1.7%	14.7% ± 12.1%
tyv	44.8% ± 3.8%	47.7% ± 2.6%	46.2% ± 5.7%	49.9% ± 2.5%	11.5% ± 5.3%	19.1% ± 9.1%
udm	57.0% ± 6.1%	60.9% ± 3.4%	51.4% ± 9.0%	58.9% ± 3.2%	31.1% ± 3.0%	30.2% ± 4.6%
ug	43.8% ± 11.5%	45.3% ± 4.4%	42.9% ± 5.1%	46.9% ± 3.7%	22.0% ± 4.4%	20.1% ± 9.2%
uk	64.6% ± 6.0%	61.5% ± 2.9%	61.4% ± 7.1%	52.8% ± 2.1%	29.9% ± 2.4%	30.5% ± 3.3%
ur	66.7% ± 6.7%	65.4% ± 4.7%	63.2% ± 6.7%	64.9% ± 3.5%	28.8% ± 2.0%	27.3% ± 3.9%
uz	61.7% ± 6.8%	57.5% ± 2.8%	50.9% ± 3.1%	56.4% ± 2.0%	27.5% ± 2.4%	27.2% ± 2.5%
vec	67.3% ± 4.0%	68.4% ± 3.0%	60.6% ± 4.0%	56.5% ± 4.2%	17.2% ± 4.6%	26.6% ± 5.7%
vep	48.6% ± 3.1%	53.0% ± 1.4%	44.7% ± 3.0%	51.3% ± 2.1%	25.6% ± 2.9%	24.8% ± 11.7%
vi	78.9% ± 5.5%	75.8% ± 4.5%	72.8% ± 7.3%	67.7% ± 3.7%	30.1% ± 3.2%	30.1% ± 2.7%
vls	70.6% ± 4.5%	69.3% ± 3.5%	63.8% ± 4.6%	58.0% ± 2.9%	21.4% ± 3.1%	26.1% ± 11.4%
vo	60.2% ± 5.4%	58.5% ± 5.6%	13.4% ± 10.9%	52.1% ± 7.5%	40.3% ± 3.9%	42.1% ± 2.6%
wa	46.9% ± 3.6%	48.1% ± 2.4%	43.4% ± 4.2%	48.7% ± 2.3%	5.6% ± 4.4%	6.1% ± 9.6%
war	48.1% ± 16.6%	58.5% ± 8.9%	32.6% ± 11.7%	48.8% ± 5.3%	53.1% ± 2.7%	59.0% ± 4.1%
wo	22.6% ± 6.8%	25.3% ± 5.0%	15.7% ± 4.8%	24.3% ± 3.6%	2.5% ± 3.8%	6.4% ± 10.2%
wuu	43.8% ± 6.9%	42.6% ± 4.7%	40.2% ± 4.0%	41.1% ± 1.8%	16.5% ± 2.9%	15.6% ± 7.8%
xh	37.0% ± 6.3%	43.4% ± 5.3%	18.3% ± 6.7%	24.1% ± 3.9%	5.1% ± 3.2%	13.5% ± 6.2%
xmf	49.7% ± 2.6%	50.0% ± 3.0%	43.9% ± 2.0%	45.1% ± 1.1%	20.1% ± 3.1%	21.5% ± 9.8%
yi	59.2% ± 6.8%	62.2% ± 2.7%	55.8% ± 5.1%	57.1% ± 2.9%	26.1% ± 3.4%	29.7% ± 4.3%
yo	39.4% ± 5.0%	45.2% ± 7.3%	42.2% ± 4.8%	51.0% ± 3.3%	0.1% ± 0.3%	16.0% ± 8.6%
yue	56.3% ± 3.5%	51.2% ± 2.6%	48.1% ± 5.1%	52.7% ± 2.3%	12.2% ± 8.4%	15.6% ± 6.9%
zea	71.8% ± 2.5%	69.5% ± 2.9%	62.8% ± 5.6%	67.0% ± 2.3%	23.3% ± 3.4%	29.1% ± 3.6%
zh-cn	54.3% ± 2.9%	47.0% ± 2.8%	51.1% ± 4.9%	49.2% ± 3.6%	8.8% ± 11.3%	16.1% ± 8.7%
zh-tw	58.1% ± 4.3%	48.7% ± 2.7%	52.4% ± 3.0%	54.9% ± 2.4%	15.3% ± 7.1%	14.3% ± 11.1%
zu	37.9% ± 5.0%	46.8% ± 4.8%	27.2% ± 4.8%	34.0% ± 2.4%	11.8% ± 6.6%	14.2% ± 11.3%

Table 7: The F1-scores on MultiWikiQA for languages 196-261, sorted alphabetically.

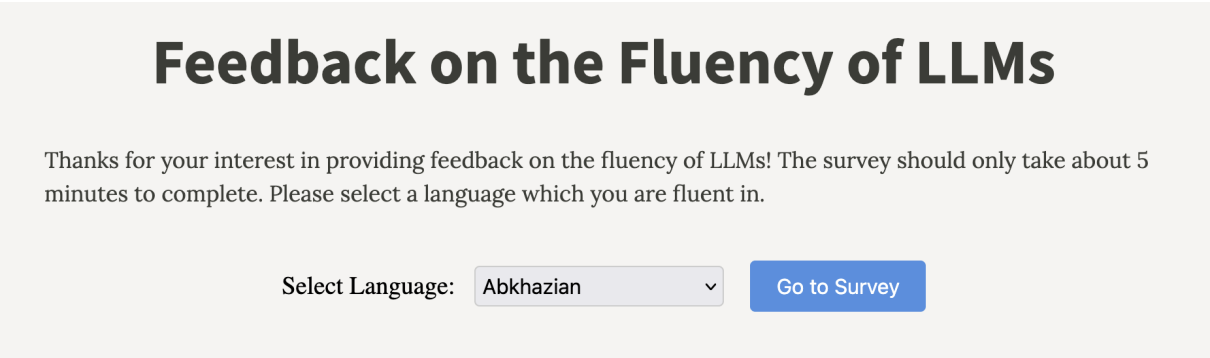

Figure 7: The survey routing interface.

```
<script lang="ts" setup>
const languageToSurveyUrl: Record<string, string> = {
  Abkhazian: "https://forms.cloud.microsoft/e/pdmAKbsRk1",
  Acehnese: "",
  (...)
};

function goToSurvey() {
  const selectElement = document.getElementById(
    "language-select",
  ) as HTMLSelectElement;
  const selectedLanguage = selectElement.value;
  const surveyUrl = languageToSurveyUrl[selectedLanguage];
  if (surveyUrl) {
    window.open(surveyUrl, "_blank");
  } else {
    window.open(
      "mailto:[redacted]@[redacted].[redacted]?" +
        "subject=Fluency%20Survey%20Language%20Support - " +
        selectedLanguage +
        "&" +
        "body=I would like to request support for " +
        selectedLanguage +
        " in the fluency survey. Thanks!",
      "_blank",
    );
  }
}
</script>
<template>
  <h1 class="centered">Feedback on the Fluency of LLMs</h1>
  <p class="centered-box serif-text">
    Thanks for your interest in providing feedback on the fluency of LLMs! The
    survey should only take about 5 minutes to complete. Please select a
    language which you are fluent in.
  </p>

  <br />
  <br />

  <div class="centered">
    <label for="language-select" class="language-label"
      >Select Language:
    </label>
    <select id="language-select" class="dropdown">
      <option
        v-for="language in Object.keys(languageToSurveyUrl)"
        v-bind:key="language"
        :value="language"
      >
        {{ language }}
      </option>
    </select>
    <button class="button" @click="goToSurvey">Go to Survey</button>
  </div>
</template>
<style scoped>
h1 {
  font-size: 3rem;
}
.centered {
  text-align: center;
}
.language-label {
  font-size: 1.2rem;
  margin-right: 10px;
}
.dropdown {
  font-size: 1rem;
  padding: 5px;
  margin-right: 20px;
  border: 1px solid #ccc;
  border-radius: 4px;
}
.button {
  font-size: 1rem;
  padding: 10px 20px;
  color: white;
  background-color: #4a90e2;
  border: none;
  border-radius: 4px;
  cursor: pointer;
}
.button:hover {
  background-color: #357abd;
}
</style>
```

Figure 8: The source code for the Vue.js survey routing interface component